

Project 2

```
library(scales)
members <- readr::read_csv('https://raw.githubusercontent.com/rfordatascience/tidytuesday/master/data/2020/2020-09-22/members.csv')
head(members)
```

```
# A tibble: 6 × 21
  expedition_id member_id peak_id peak_name year season sex age
<chr>          <chr>    <chr>    <chr>    <dbl> <chr> <chr> <dbl>
<chr>
1 AMAD78301     AMAD7830... AMAD     Ama Dabl... 1978 Autumn M      40
France
2 AMAD78301     AMAD7830... AMAD     Ama Dabl... 1978 Autumn M      41
France
3 AMAD78301     AMAD7830... AMAD     Ama Dabl... 1978 Autumn M      27
France
4 AMAD78301     AMAD7830... AMAD     Ama Dabl... 1978 Autumn M      40
France
5 AMAD78301     AMAD7830... AMAD     Ama Dabl... 1978 Autumn M      34
France
6 AMAD78301     AMAD7830... AMAD     Ama Dabl... 1978 Autumn M      25
France
# i 12 more variables: expedition_role <chr>, hired <lgl>,
#   highpoint_metres <dbl>, success <lgl>, solo <lgl>, oxygen_used <lgl>,
#   died <lgl>, death_cause <chr>, death_height_metres <dbl>, injured <lgl>,
#   injury_type <chr>, injury_height_metres <dbl>
```

Questions for the Analysis

Question 1: Does a climber's age influence their likelihood of success and survival, and how do these rates compare across different age groups?

Question 2: How have success and death rates evolved over time for the five most climbed peaks, and how does the usage of supplemental oxygen compare across the peaks?

Introduction

For this project, we will be working on the data related to Himalayan Expedition. The data that we will be using is publicly available on the [Himalayan Database website](#) and detailed descriptions of the tables and columns can be found in [this github repository](#). For this project, we will be particularly focusing on the `members` table. The dataset contains the records, from Summer 1905 to Spring 2019, of the participants who went on expeditions of climbing the peaks of the Himalayan ranges in Nepal. It captures data on the participant, their role, the peak they climbed, when they climbed, how far they climbed, whether they suffered any injuries and whether they survived the climb.

Here are the 2 questions that we came up for the analysis:

- 1) Does a climber's age influence their likelihood of success and survival, and how do these rates compare across different age groups?
 - This will help us understand whether a certain age groups of climbers have a higher likelihood of success of survival compared to the others.
 - This requires us to compare the success rates (rate of successful climbs), death rates and injury rates among different age groups of climbers.
 - We will be working with the following columns to answer this question:
 - **age**: the age of the climber in years. The age of the climbers range from 7 to 85 years.
 - **success**: whether the person was successful in climbing a main peak or sub-peak, depending on the goal of expedition. The column contains binary (TRUE, FALSE) values.
 - **died**: Whether the person died. The column contains binary (TRUE, FALSE) values.
 - **injured**: Whether the person was injured. The column contains binary (TRUE, FALSE) values.
- 2) How have success and death rates evolved over time for the five most climbed peaks, and how does the usage of supplemental oxygen compare across the peaks?
 - This will tell us whether the success rate (rate of successful climbs) and death rate has increased, decreased or stayed the same over the years for the five most climbed peaks.
 - We will be working with the following columns to answer this question:
 - **peak_id**: Unique identifier for peak.
 - **peak_name**: Common name for peak.
 - **year**: Year of expedition.
 - **success**: whether the person was successful in climbing a main peak or sub-peak, depending on the goal of expedition. The column contains binary (TRUE, FALSE) values.
 - **died**: Whether the person died. The column contains binary (TRUE, FALSE) values.
 - **oxygen_used**: Whether the climber used oxygen. The column contains binary (TRUE, FALSE) values.

Approach

Approach to answer Question 1:

To analyze how age impacts climbing outcomes, we will first filter out records with missing age values. The age variable will then be binned into six distinct age groups: 20, 20–29, 30–39, 40–49, 50–59, and 60+. For each age group, we will calculate the mean `success_rate`, `death_rate`, and `injury_rate`. Finally, the data will be pivoted into a long format; we will create 2 columns - `metric` (to store the metric name - `success_rate`, `death_rate` or `injury_rate`) and `rate` (to store the values of the corresponding metric).

Choice of visualization:

We will use the **bar chart** visualization, with age group on the x-axis and the metric rate on the y-axis. The bar chart will allow a direct comparison of metrics across categorical age groups. We will facet by metric using independent y-axis scales to ensure that small-magnitude data (like death rates) remain visible alongside high-magnitude data (like success rates). A custom sequential color palette will be applied to the age groups to reflect the ordinal nature of aging. Values will be displayed as percentages rather than decimals to improve readability (since metrics such as death rate represent very small values, hence, the extra leading zeros in decimals can make it harder to quickly distinguish the magnitude of the difference).

Approach to answer Question 2:

We will first filter the data of the top five most climbed peaks. The data will then be aggregated by `peak_name` and `year` to calculate annual averages for successful climbs, deaths, and oxygen usage. To ensure a clean trend analysis, we will exclude 2019 data (due to incomplete seasonal data). The resulting dataset will then be pivoted to a longer format by creating separate columns for the calculated metrics and their respective values.

Choice of visualization:

We will be using a **faceted line chart** to visualize the data by peak and metric. Faceting by both peak and metric will reduce visual clutter. We will overlay a LOESS smoothed trend line over the raw annual fluctuations. The years will be sectioned into 4 eras - Pioneer (1905-1970), Expansion (1971-1990), Commercial (1991-2010) and Modern (2011-2019). We will be highlighting the eras in the background in each panel to show how the metrics vary across the eras. Again, we will be representing the rates as percentages instead of decimals.

Analysis

Visualization for Question 1

```
age_data <- members |>
  filter(!is.na(age)) |> # filtering out the NA rows from the age column
  mutate(
    age_group = case_when( # creating age groups
      age < 20 ~ "<20",
      age >= 20 & age < 30 ~ "20-29",
      age >= 30 & age < 40 ~ "30-39",
      age >= 40 & age < 50 ~ "40-49",
      age >= 50 & age < 60 ~ "50-59",
      age >= 60 ~ "60+"
    )
  ) |>
  mutate(age_group = factor(age_group, levels = rev(c("<20", "20-29", "30-39",
"40-49", "50-59", "60+")))) |>
  group_by(age_group) |> # grouping the data by age_groups
  summarize(
    success_rate = mean(success, na.rm = TRUE), # calculating the average
number of successful climbs per age group
    death_rate = mean(died, na.rm = TRUE), # calculating the average number of
deaths per age group
    injury_rate = mean(injured, na.rm = TRUE), # calculating the average
number of injuries per age group
    .groups = "drop"
  ) |>
  # pivoting the data and creating a new columns :
  # `metric` to store the metric name (success_rate, death_rate or
injury_rate)
  # `rate` to store the values of the corresponding metric
  pivot_longer( # converting columns to rows
    cols = c(success_rate, death_rate, injury_rate),
    names_to = "metric",
    values_to = "rate"
  )

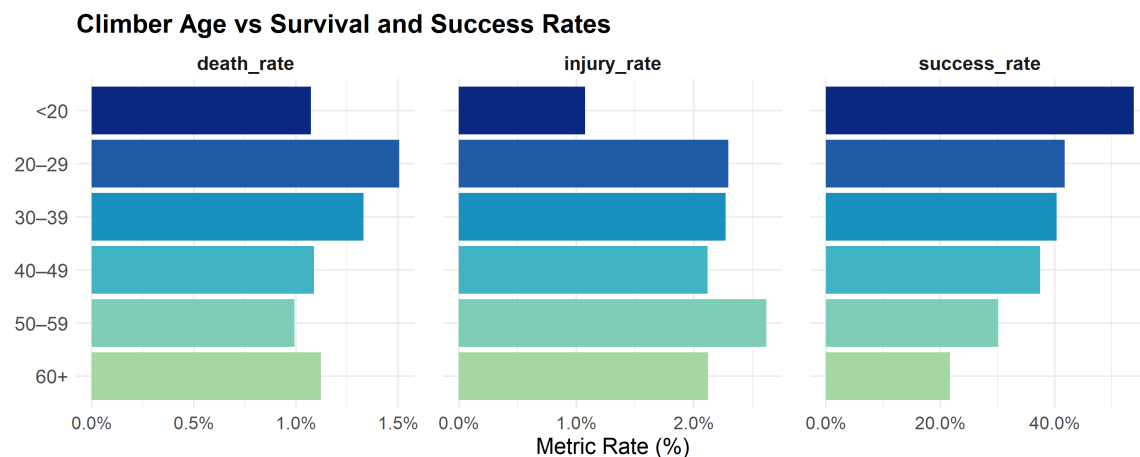
# Creating custom colour pallet
himalayan_pastels <- c("<20" = "#0c2c84", "20-29" = "#225ea8", "30-39" =
"#1d91c0",
                      "40-49" = "#41b6c4", "50-59" = "#7fcdbb", "60+" =
"#a7d9a4")

# Plotting a bar chart
ggplot(age_data, aes(y = age_group, x = rate, fill = age_group)) +
  geom_col() +
  facet_wrap(~ metric, # faceting by the metric type
```

```

    scales = "free_x") + # different scales for each panel/facet
# Giving the plot a title and axis labels
labs(
  title = "Climber Age vs Survival and Success Rates",
  y = NULL,
  x = "Metric Rate (%)"
) +
# Representing the rates in the form of percentages
scale_x_continuous(labels = scales::label_percent(accuracy = 0.1)) +
scale_fill_manual(values = himalayan_pastels) +
theme_minimal() +
theme(
  legend.position = "none", # removing the legend
  strip.text.x = element_text(face = "bold", size = 12), # making the panel
titles bold
  axis.title.x = element_text(size = 14), # increase x-axis label size
  panel.spacing = unit(1.25, "lines"), # adding some spaces between the
panels
  plot.title = element_text(face = "bold", size = 16), # making the plot
title bold
  axis.text.y = element_text(size = 12), # increasing the size of the x and
y axis tick labels
  axis.text.x = element_text(size = 11)
)

```



NOTE: Although rates are computed per age group, the number of observations varies significantly across the groups. Smaller groups (e.g., <20 and 60+) may exhibit higher variability and less reliable estimates compared to larger groups.

```

# Counting the number of records under each age group
age_data <- members |>
  filter(!is.na(age)) |> # filtering out the NA rows from the age column

```

```

mutate(
  age_group = case_when( # creating age groups
    age < 20 ~ "<20",
    age >= 20 & age < 30 ~ "20-29",
    age >= 30 & age < 40 ~ "30-39",
    age >= 40 & age < 50 ~ "40-49",
    age >= 50 & age < 60 ~ "50-59",
    age >= 60 ~ "60+"
  )
) |>
count(age_group) |>
arrange(n)

glimpse(age_data)

```

```

Rows: 6
Columns: 2
$ age_group <chr> "<20", "60+", "50-59", "40-49", "20-29", "30-39"
$ n           <int> 559, 2407, 7566, 17201, 18070, 27219

```

Visualization for Question 2

```

# Defining sections of each era for background shading
# Refer to NOTE*
era_definitions <- data.frame(
  xmin = c(-Inf, 1970.5, 1990.5, 2010.5), # lower limit of each era (start
year of era)
  xmax = c(1970.5, 1990.5, 2010.5, Inf), # upper limit of each era (end year
of era)
  # the years are represented as decimals so that the transitions don't
overlap with the datapoints.
  era = factor(c("Pioneer", "Expansion", "Commercial", "Modern"),
    levels = c("Pioneer", "Expansion", "Commercial", "Modern")), #
defining the order
  fill = c("#D1E8FF", "#A0E4F1", "#C2F0E5", "#D9F5D6") # Soft colors for
background
)

# getting the top 5 most climbed peaks
top_5_peaks <- members |>
  group_by(peak_id) |> # grouping by the peak id
  summarize(
    num_climbs = n() # getting the number of expeditions per peak
  ) |>
  arrange(desc(num_climbs)) |> # sorting in decreasing order of the count of

```

```

expeditions
  slice(1:5) |> # slicing the top 5 peaks
  select(peak_id) # retaining only the peak_id column

# Data Wrangling - preparing the data for the visualization
trend_data <- members |>
  # filtering data of top 5 most climbed peaks
  filter(peak_id %in% top_5_peaks$peak_id &
    # excluding 2019 data (as we only have data till Spring and this
    may affect our trend line)
    year < 2019) |>
  filter(!(peak_name == "Everest" & year == 2015)) |> # Refer to NOTE**
  group_by(peak_name, year) |> # grouping by the peaks and the year
  summarize( #calculating the average success, deaths and usage of oxygen each
year for each peak
    success_rate = mean(success, na.rm = TRUE),
    death_rate = mean(died, na.rm = TRUE),
    oxygen_usage = mean(oxygen_used, na.rm = TRUE),
    .groups = "drop"
  ) |>
  # reordering the data
  mutate(peak_name = fct_reorder(peak_name, oxygen_usage, .desc = TRUE)) |>
  # pivoting the data and creating a new columns :
  # `metric` to store the metric name (success_rate, death_rate or
injury_rate)
  # `rate` to store the values of the corresponding metric
  pivot_longer( # converting columns to rows
    cols = c(success_rate, death_rate, oxygen_usage),
    names_to = "metric",
    values_to = "rate"
  )

# Creating the Visualization
ggplot(trend_data, aes(x = year, y = rate)) +
  # Applying background shading for each era
  geom_rect(data = era_definitions,
    aes(xmin = xmin, xmax = xmax, ymin = -Inf, ymax = Inf, fill =
era),
    alpha = 0.15, inherit.aes = FALSE) +

  # Plotting the raw data trend line in the background
  geom_line(aes(group = peak_name), color = "grey70", alpha = 0.6) +

  # Plotting the smoothened trend line
  geom_smooth(aes(color = peak_name), method = "loess", se = FALSE,
    span = 0.5, linewidth = 1) +

```

```

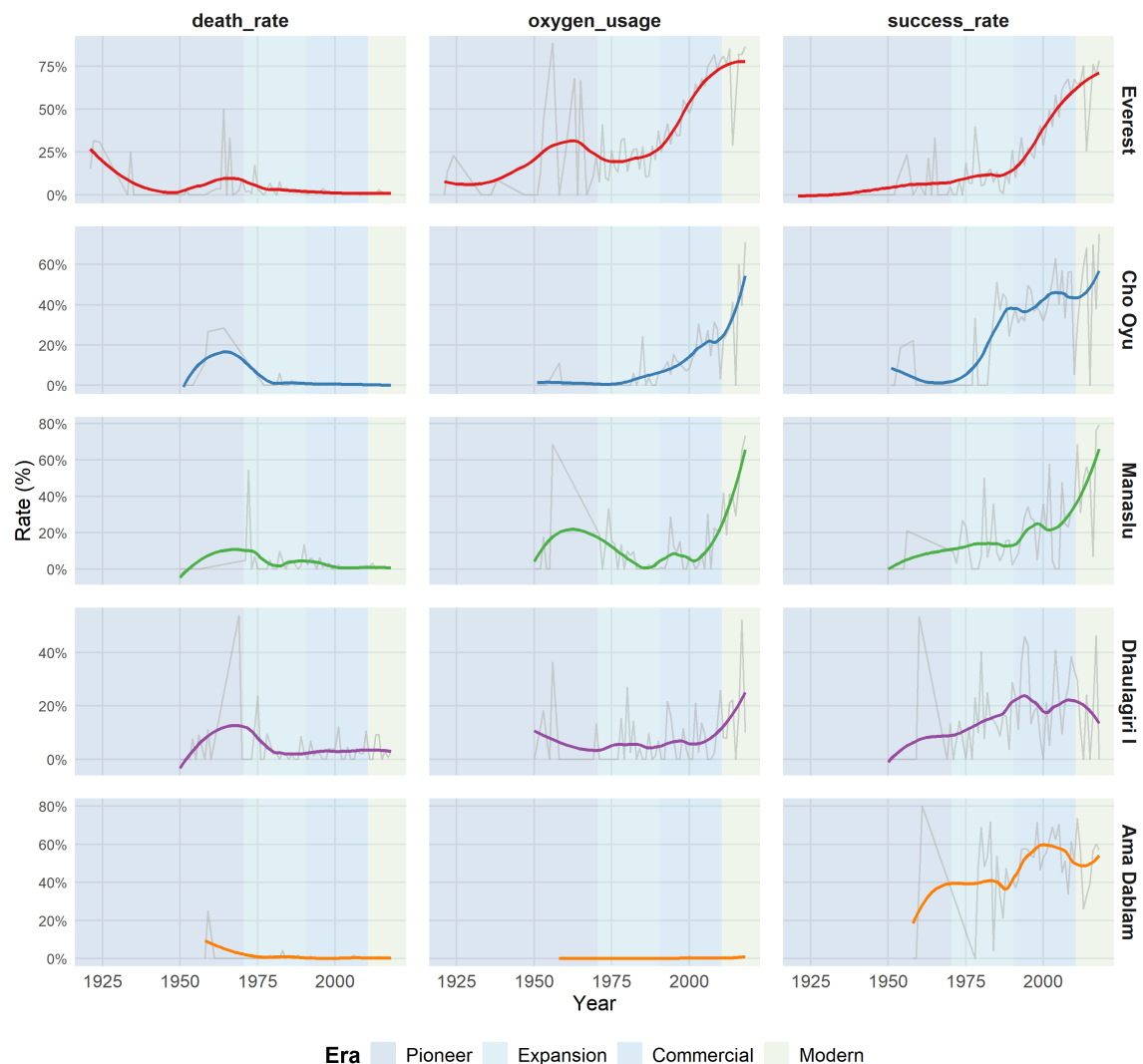
# Faceting by Peak and Metric
# This creates a grid where rows = peaks and columns = metrics
facet_grid(peak_name ~ metric, scales = "free_y") +

# Styling
scale_y_continuous(labels = label_percent()) + # Representing the data values
in the form of percentages
scale_fill_manual(values = c("Pioneer" = "#225ea8", "Expansion" =
"#41b6c4",
                                "Commercial" = "#1d91c0", "Modern" =
"#98c985")) +
scale_color_brewer(palette = "Set1", guide = "none") + # colours for the
trend lines
theme_minimal() +
labs( # adding a title and axis labels
  title = "Himalayan Expedition Outcome Trends among the Top 5 Most Climbed
Peaks",
  subtitle = "Comparison of metrics across historical eras of each peak",
  x = "Year",
  y = "Rate (%)",
  fill = "Era"
) +
theme( # Formatting text and graph elements
  strip.text.y = element_text(face = "bold", size = 12), # Peak names
horizontal
  strip.text.x = element_text(face = "bold", size = 12),
  axis.title.y = element_text(size = 13), # increase x and y-axis title
sizes
  axis.title.x = element_text(size = 13),
  legend.position = "bottom",
  panel.grid.minor = element_blank(),
  panel.spacing = unit(1, "lines"), # adding some spaces between the panels
  plot.title = element_text(face = "bold", size = 16),
  axis.text.x = element_text(size = 11),
  legend.text = element_text(size = 12), # changing legend text size
  legend.title = element_text(size = 14, face = "bold")
)

```


Himalayan Expedition Outcome Trends among the Top 5 Most Climbed Peaks

Comparison of metrics across historical eras of each peak



NOTE* : The reason why we are binning the years column is since we have data over 114 years, it would be difficult to visualize the trend over the years. The data is sectioned into 4 eras - Pioneer (1905-1970), Expansion (1971-1990), Commercial (1991-2010) and Modern (2011-2019). The reason we bin the years into eras rather than decades is because there is still a massive difference in the number of records under each decade. The number of records are comparatively more balanced after binning the data into eras.

Here are the number of expeditions under each decade and era:

```
# Number of expeditions under each decade
members |>
  filter(peak_id %in% top_5_peaks$peak_id) |>
```

```
mutate(
  decade = case_when(
    year < 1910 ~ "< 1910",
    year >= 1910 & year < 1920 ~ "1910-1919",
    year >= 1920 & year < 1930 ~ "1920-1929",
    year >= 1930 & year < 1940 ~ "1930-1939",
    year >= 1940 & year < 1950 ~ "1940-1949",
    year >= 1950 & year < 1960 ~ "1950-1959",
    year >= 1960 & year < 1970 ~ "1960-1969",
    year >= 1970 & year < 1980 ~ "1970-1979",
    year >= 1980 & year < 1990 ~ "1980-1989",
    year >= 1990 & year < 2000 ~ "1990-1999",
    year >= 2000 & year < 2010 ~ "2000-2009",
    year >= 2010 ~ "2009+"
  )
) |>
group_by(decade) |>
summarize(
  counts = n()
) |>
glimpse()
```

```
Rows: 10
Columns: 2
$ decade <chr> "1920-1929", "1930-1939", "1940-1949", "1950-1959", "1960-1969"...
$ counts <int> 48, 121, 3, 304, 210, 1098, 3842, 8495, 14752, 17421
```

```
# Number of expeditions under each era
members |>
filter(peak_id %in% top_5_peaks$peak_id) |>
mutate(
  era = cut(year,
    breaks = c(-Inf, 1970, 1990, 2010, Inf),
    labels = c("Pioneer", "Expansion", "Commercial", "Modern"))
) |>
count(era) |> glimpse()
```

```
Rows: 4
Columns: 2
$ era <fct> Pioneer, Expansion, Commercial, Modern
$ n <int> 784, 5468, 24352, 15690
```

NOTE** : In 2015, a massive earthquake in Nepal caused an avalanche that hit Everest Base Camp, leading to the complete cancellation of the climbing season that year. This

information reflects in the data as during this year, records of peak Everest had 0 successful climb and 0 use of oxygen (we see a sudden drop in the values to 0 which is contrary to the consistent upward trend in the success rate and oxygen usage that we see otherwise through the years in the Commercial and Modern eras).

```
# Checking the success, death and oxygen usage rates for Everest peak from
2010 - 2018
members |>
  filter(peak_id %in% top_5_peaks$peak_id & year < 2019) |>
  filter(peak_name == "Everest" & year > 2009) |>
  group_by(peak_name, year) |>
  summarize(
    success_rate = round(mean(success, na.rm = TRUE),3),
    death_rate = round(mean(died, na.rm = TRUE),3),
    oxygen_usage = round(mean(oxygen_used, na.rm = TRUE),3),
    .groups = "drop"
  ) |>
  glimpse()
```

```
Rows: 9
Columns: 5
$ peak_name    <chr> "Everest", "Everest", "Everest", "Everest", "Everest",
"E...
$ year         <dbl> 2010, 2011, 2012, 2013, 2014, 2015, 2016, 2017, 2018
$ success_rate <dbl> 0.673, 0.646, 0.657, 0.752, 0.256, 0.000, 0.761, 0.701,
0...
$ death_rate   <dbl> 0.004, 0.005, 0.011, 0.009, 0.033, 0.015, 0.006, 0.006,
0...
$ oxygen_usage <dbl> 0.786, 0.809, 0.765, 0.853, 0.291, 0.000, 0.821, 0.819,
0...
```

Discussion

Question 1 visualization insights:

1. There is a clear downward trend in success rates as age increases. Climbers in the younger age groups (<20 and 20–29) have the highest success rates. The 60+ age group has the lowest success rate.
2. The 20–29 age group has the highest death rate. Death rates show a general decrease with age, with a slight increase in the 60+ group. This might suggest that younger climbers take more risks, or attempt more dangerous routes compared to older, more cautious climbers.
3. The 50-59 age group has the highest injury rate.
4. We also notice a trade-off between success and safety. Younger climbers observe high success rates, but at the same time, high death rates too. Older climbers observe low success rates but low death rates as well.
5. The 60+ category of climbers have the lowest success rate but at the same time their death_rate is lower than the 20–29 and 30–39 groups. This supports the idea that the oldest climbers are more cautious and avoid taking risk.

To conclude, yes, a climber's age does influence their likelihood of success and survival.

Question 2 visualization insights:

1. Everest, Cho Oyu, Manaslu, Dhaulagiri I and Ama Dabla are the top five most climbed peaks.
2. Death rates have declined significantly over time. For Everest, Cho Oyu, and Manaslu, death rates decrease steadily and approach near-zero levels in the Commercial and Modern eras. Ama Dablam shows near-zero death rates from the Expansion era onwards. Dhaulagiri I is an exception, with a slight increase in later years, though still lower than early-era levels. Overall, death rates are substantially lower in the Commercial and Modern eras compared to the Pioneer and Expansion eras.
3. Everest, Cho Oyu, and Manaslu exhibit a strong upward trend in success rates. Dhaulagiri I and Ama Dablam show more variability and less consistent improvement. In fact, Dhaulagiri I shows a decline in the success rate in later years. But on a general note, the success rates are higher in the Commercial and Modern eras compared to the Pioneer and Expansion eras.
4. Oxygen usage varies by peak. The rate of oxygen usage is significantly higher for Everest, Cho Oyu and Manaslu compared to the other 2 peaks. Ama Dablam has near-zero oxygen usage throughout, indicating that supplemental oxygen is rarely used. Dhaulagiri I shows a gradual increase in oxygen usage, especially in the Modern era. If we were to look at the heights of these mountains, Everest (8,849m), Cho

Oyu (8,188m), Manaslu (8,163m) and Dhaulagiri (8,167m) have heights greater than 8000 meters, whereas, Ama Dablam has a height of approximately 6800 meters. This highlights a pattern that higher altitude peaks require greater oxygen usage.

Generative AI Disclosure

Made use of Generative AI ONLY to search for the base syntax of the ggplot functions in R and to help come up with the mountainous-themed color palettes for the visualizations.